

Java Script - Day-4 - Hands on -Gurunath Rageni

Problem 1

Assessment Goal: Ensure learners understand responsiveness and screen adaptability.

Hands-on Tasks:

1. Add viewport meta tag to the HTML page
2. Use media queries to:
 - o Change background color on mobile screen
 - o Adjust font size for smaller screens
3. Convert navigation into vertical layout on mobile
4. Test the page using browser responsive mode

Code:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <!-- Viewport for responsiveness -->
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Responsive Flexbox Example</title>
  <link rel="stylesheet" href="styles.css">
</head>
<body>
  <header>
    <h1>My Responsive Website</h1>
  </header>
  <nav>
    <a href="#">Home</a>
    <a href="#">Courses</a>
    <a href="#">About</a>
    <a href="#">Contact</a>
  </nav>
```

```
<section>

  <p>

    This layout uses Flexbox for professional responsive design.

    Resize the browser to see changes.

  </p>

</section>
```

```
</body>
</html>
```

```
/* Desktop Styles (Default) */
```

```
body {
  margin: 0;
  font-family: Arial, sans-serif;
  background-color: lightblue;
  text-align: center;
}
```

```
header {
  background-color: rgb(30, 30, 32);
  color: white;
  padding: 20px;
}
```

```
/* Flexbox Navigation */
```

```
nav {
  display: flex;      /* Enable flexbox */
  justify-content: center; /* Center horizontally */
  gap: 20px;          /* Space between items */
  background-color: rgb(139, 149, 155);
  padding: 15px;
```

```
}
```

```
nav a {  
    color: white;  
    text-decoration: none;  
    font-size: 20px;  
}
```

```
section {  
    padding: 20px;  
    font-size: 18px;  
}
```

```
/* Mobile Styles */
```

```
@media (max-width: 600px) {
```

```
    /* Change background */  
    body {  
        background-color: rgb(206, 74, 18);  
    }
```

```
    /* Adjust font size */  
    header h1 {  
        font-size: 22px;  
    }
```

```
    section {  
        font-size: 14px;  
    }
```

```

/* Convert to vertical layout */

nav{

    flex-direction:column;

    align-items: center; /* Center items */

}

}

```

Code Screenshots:

```

WEEK_1 > Day4_handson_js > Problem1 > index.html > html > body > nav
1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4      <meta charset="UTF-8">
5      <!-- Viewport for responsiveness -->
6      <meta name="viewport" content="width=device-width, initial-scale=1.0">
7      <title>Responsive Flexbox Example</title>
8      <link rel="stylesheet" href="styles.css">
9  </head>
10 <body>
11     <header>
12         <h1>My Responsive Website</h1>
13     </header>
14     <nav>
15         <a href="#">Home</a>
16         <a href="#">Courses</a>
17         <a href="#">About</a>
18         <a href="#">Contact</a>
19     </nav>
20     <section>
21         <p>
22             This layout uses Flexbox for professional responsive design.
23             Resize the browser to see changes.
24         </p>
25     </section>
26
27 </body>
28 </html>

```

WEEK_1 > Day4_handson_js > Problem1 > # styles.css > nav

```
1  /* Desktop Styles (Default) */
2  body {
3      margin: 0;
4      font-family: Arial, sans-serif;
5      background-color: lightblue;
6      text-align: center;
7  }
8
9  header {
10     background-color: rgb(30, 30, 32);
11     color: white;
12     padding: 20px;
13 }
14
15 /* Flexbox Navigation */
16 nav {
17     display: flex;                /* Enable flexbox */
18     justify-content: center;      /* Center horizontally */
19     gap: 20px;                   /* Space between items */
20     background-color: rgb(139, 149, 155);
21     padding: 15px;
22 }
23
24 nav a {
25     color: white;
26     text-decoration: none;
27     font-size: 20px;
28 }
29
```

```

28     }
29
30     section {
31         padding: 20px;
32         font-size: 18px;
33     }
34
35
36     /* Mobile Styles */
37
38     @media (max-width: 600px) {
39
40         /* Change background */
41         body {
42             background-color:  rgb(206, 74, 18);
43         }
44
45         /* Adjust font size */
46         header h1 {
47             font-size: 22px;
48         }
49
50         section {
51             font-size: 14px;
52         }
53
54         /* Convert to vertical layout */
55         nav {

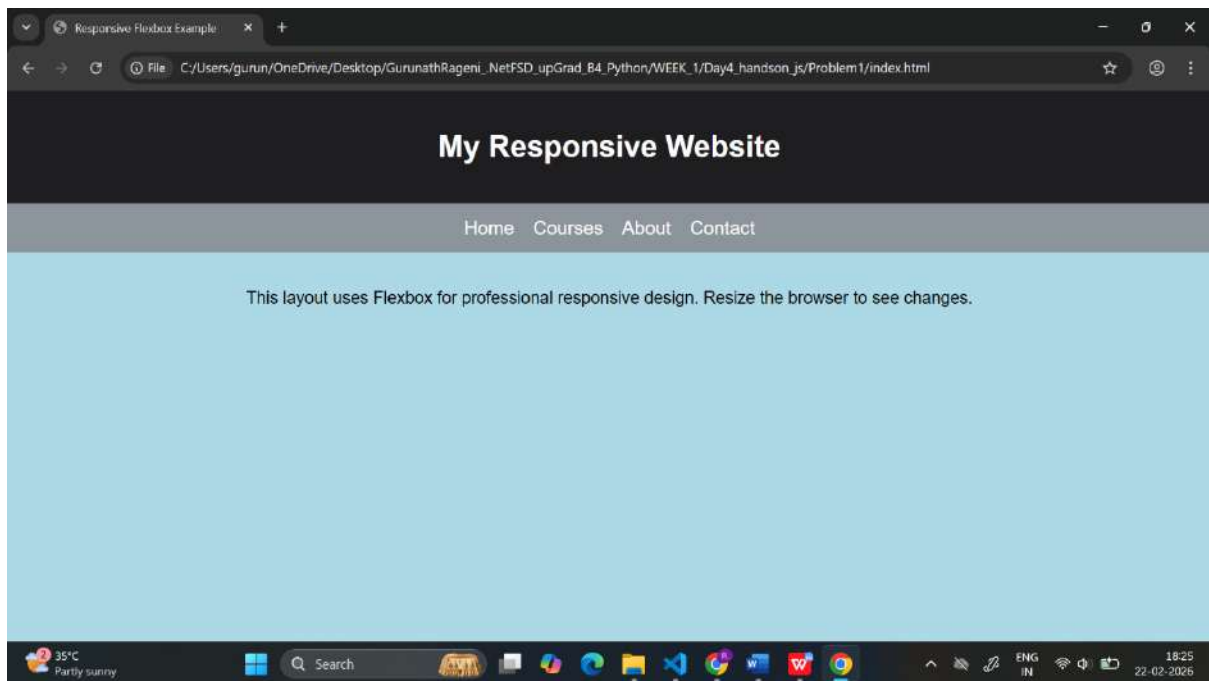
```

```

55             nav {
56                 flex-direction: column;
57                 align-items: center;          /* Center items */
58             }
59         }
60

```

Output Screenshot:



Explanation:

In this Problem, I created a Responsive Website layout using HTML and CSS with Flexbox for navigation design. The default (desktop) layout displays the navigation links horizontally using `display: flex`, `justify-content`, and `gap` for proper alignment and spacing. I implemented a media query for screens smaller than 600px to apply mobile-specific styles such as changing the background color, reducing font sizes, and converting the navigation menu into a vertical layout using `flex-direction: column`. The viewport meta tag ensures proper scaling on different devices. This project demonstrates the use of Flexbox, media queries, and mobile-responsive design principles to build an adaptive webpage layout.

Problem 2: Student Grade Evaluator (Level-1)

Scenario

A school wants a simple JavaScript program to evaluate a student's performance based on marks obtained in a subject.

📌 Requirements

- **Accept the student's marks as a variable**
- **Use if-else statements to assign grades:**
 - **Marks ≥ 75 $\rightarrow$ Grade A**
 - **Marks ≥ 60 $\rightarrow$ Grade B**
 - **Marks ≥ 40 $\rightarrow$ Grade C**
 - **Marks < 40 $\rightarrow$ Fail**

Display the grade on the web page or console

Technical Constraints

- Use JavaScript variables (let or const)
- Use numeric data types
- Use comparison and logical operators
- No functions or arrays allowed
- Output using console.log() or document.write()

Code:

```
<!DOCTYPE html>
<html lang="en">
  <head>
    <title>Marks</title></head>
  <body>
    <h1>Student Grades</h1>
    <script>
      let marks=Number(prompt("Enter Marks"))
      while(isNaN(marks)){
        marks=prompt("Enter marks Number only");
      }
      while(marks<0){
        marks=prompt("Enter positive marks only")
      }
      if(marks>=75){
        document.write("Grade A");
      }
      else if (marks >= 60) {
        document.write("Grade B");
      }
      else if (marks >= 40) {
```

```

        document.write("Grade C");
    }
    else {
        document.write("Fail");}
</script>
</body>

</html>

```

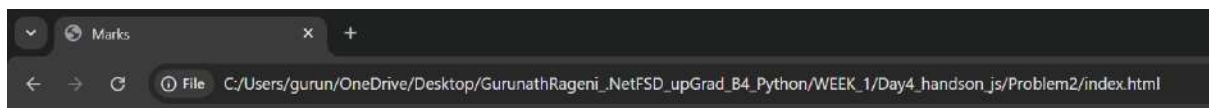
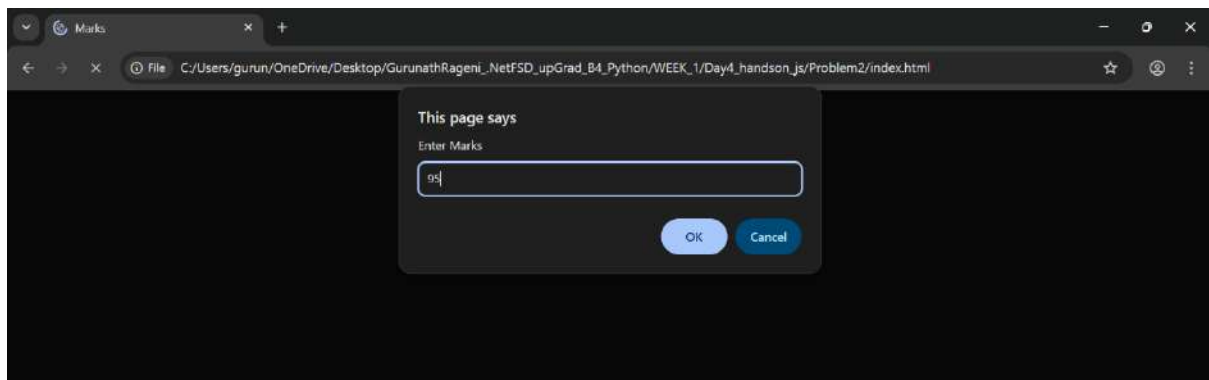
Code Screenshot:

```

<> index.html ...\Problem1 M    <> index.html ...\Problem2 M X
WEEK_1 > Day4_handson_js > Problem2 > <> index.html > html > body > script
1  <!DOCTYPE html>
2  <html lang="en">
3      <head>
4          <title>Marks</title></head>
5      <body>
6          <h1>Student Grades</h1>
7          <script>
8              let marks=Number(prompt("Enter Marks"))
9              while(isNaN(marks)){
10                 marks=prompt("Enter marks Number only");
11             }
12             while(marks<0){
13                 marks=prompt("Enter positive marks only")
14             }
15             if(marks>=75){
16                 document.write("Grade A");
17             }
18             else if (marks >= 60) {
19                 document.write("Grade B");
20             }
21             else if (marks >= 40) {
22                 document.write("Grade C");
23             }
24             else {
25                 document.write("Fail");}
26         </script>
27     </body>
28
29 </html>

```

Output Screenshot:



Explanation:

In this Problem, I created a simple Student Grades webpage using HTML and JavaScript to calculate grades based on user input. The program collects marks using the `prompt()` function and converts the input into a number using `Number()`. I implemented validation using `isNaN()` and while loops to ensure the user enters numeric and positive values only. Based on the entered marks, conditional statements (if-else if-else) are used to determine the appropriate grade (A, B, C, or Fail). This project demonstrates basic input handling, validation, and decision-making logic using JavaScript.

Problem 3: Simple Discount Calculator (Level-1)

Scenario

An online store wants to apply a discount based on the total purchase amount.

Requirements

- Store purchase amount in a variable
- Apply discount rules:
 - Amount $\geq 5000 \rightarrow$ 20% discount
 - Amount $\geq 3000 \rightarrow$ 10% discount
 - Amount $< 3000 \rightarrow$ No discount

- Calculate and display:
 - Discount amount
 - Final payable amount

Technical Constraints

- Use arithmetic operators
- Use if-else statements
- Use only primitive data types

No user input (hardcoded values allowed)

Code:

```
<!DOCTYPE html>

<html>

<head>

  <title>Simple Discount Calculator</title>

</head>

<body>

  <h1>Discount Calculator</h1>
```

```
<script>

  let amount =5000

  let discount = 0;

  if (amount >= 5000) {

    discount = amount * 0.20;

  }else if (amount >= 3000) {

    discount = amount * 0.10;

  }else {

    discount = 0;

  }

  console.log("Purchase Amount: " + amount);

  console.log("Discount: " + discount);
```

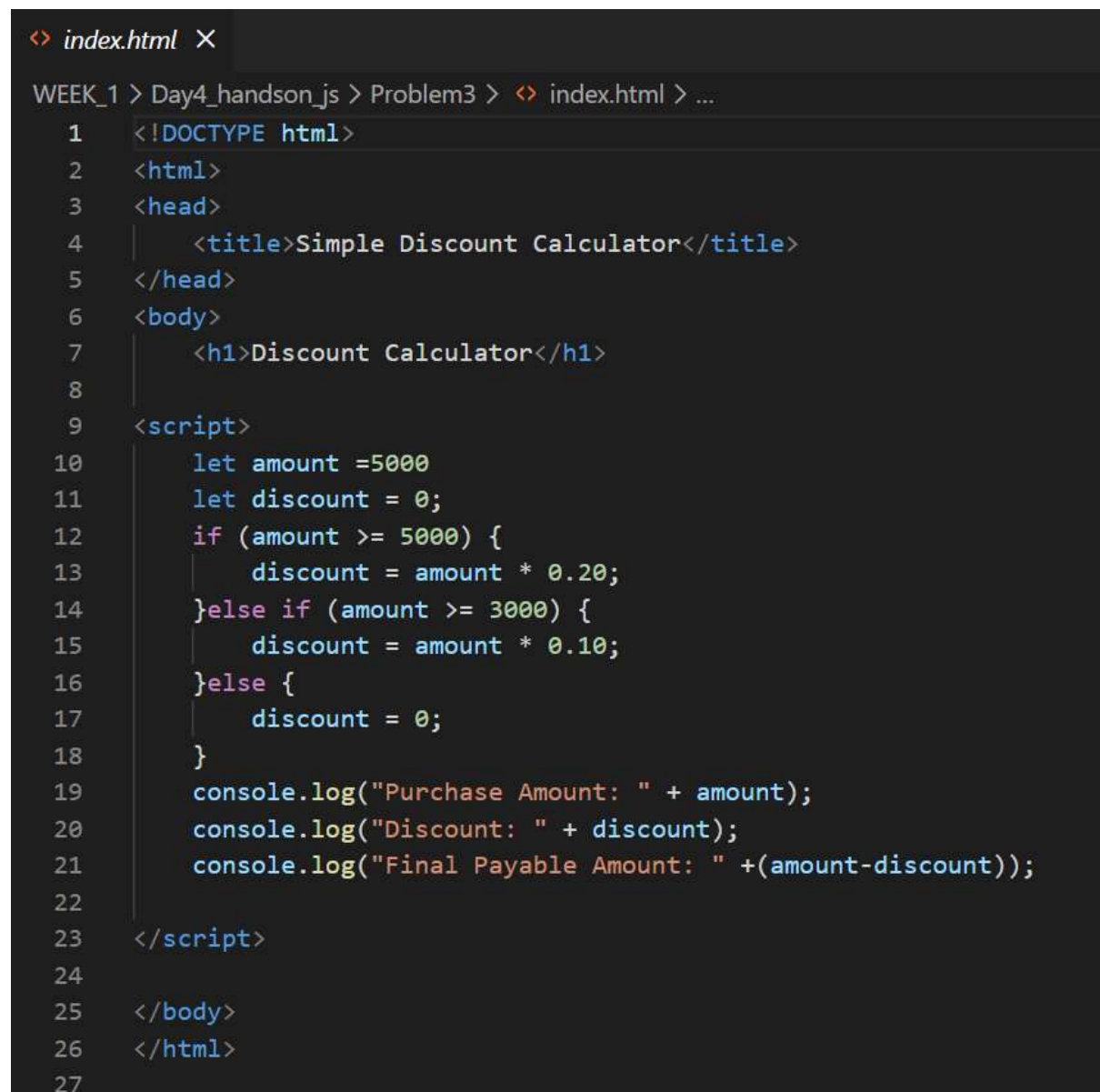
```
console.log("Final Payable Amount: " +(amount-discount));
```

```
</script>
```

```
</body>
```

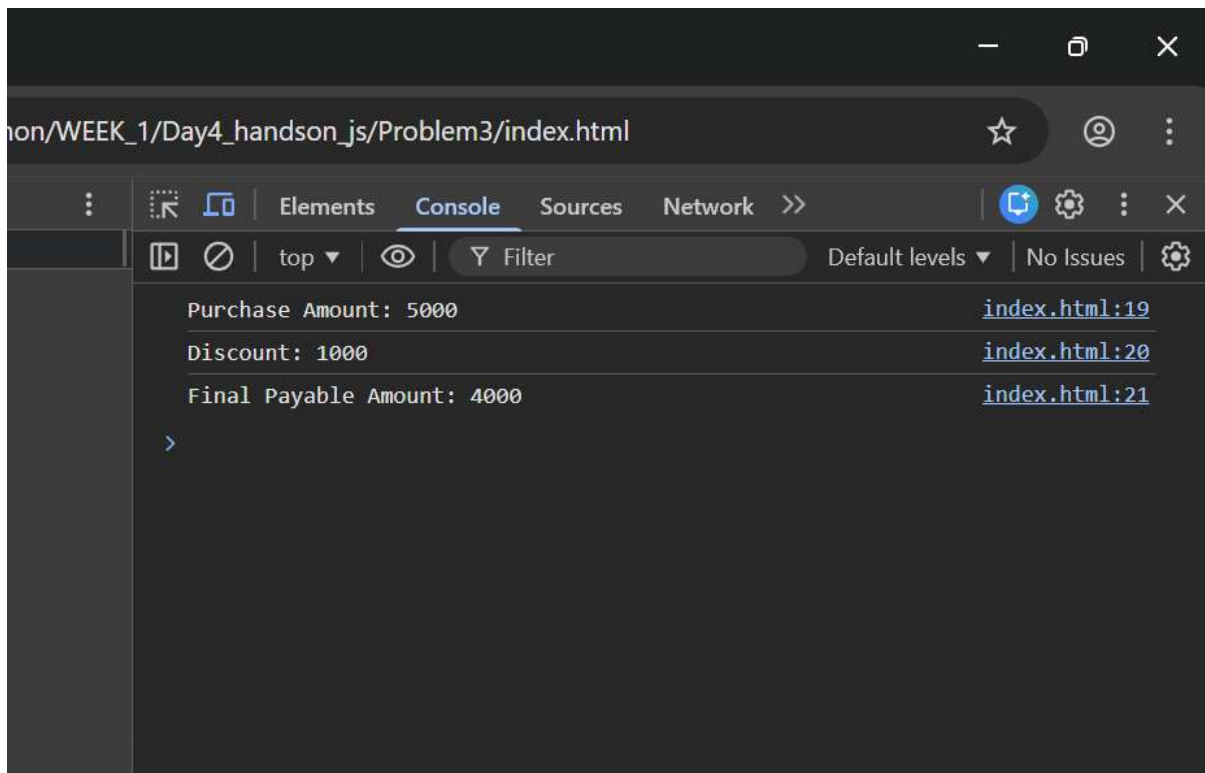
```
</html>
```

Code Screenshot:



```
<> index.html X
WEEK_1 > Day4_handson_js > Problem3 > <> index.html > ...
1  <!DOCTYPE html>
2  <html>
3  <head>
4  |   <title>Simple Discount Calculator</title>
5  </head>
6  <body>
7  |   <h1>Discount Calculator</h1>
8  |
9  <script>
10 |   let amount =5000
11 |   let discount = 0;
12 |   if (amount >= 5000) {
13 |       discount = amount * 0.20;
14 |   }else if (amount >= 3000) {
15 |       discount = amount * 0.10;
16 |   }else {
17 |       discount = 0;
18 |   }
19 |   console.log("Purchase Amount: " + amount);
20 |   console.log("Discount: " + discount);
21 |   console.log("Final Payable Amount: " +(amount-discount));
22 |
23 </script>
24
25 </body>
26 </html>
27
```

Output Screenshot:



Explanation:

In this Problem I created a Simple Discount Calculator using HTML and JavaScript to calculate the discount based on the purchase amount. The program assigns a fixed amount and uses conditional statements (if-else if-else) to apply different discount percentages depending on the value. If the amount is ₹5000 or more, a 20% discount is applied; if it is ₹3000 or more, a 10% discount is applied; otherwise, no discount is given. The final payable amount is calculated by subtracting the discount from the original amount. The results are displayed in the console, demonstrating basic arithmetic operations and decision-making logic in JavaScript.

Problem 4: Traffic Signal Simulator (Level-2)

Scenario

A traffic control system needs a JavaScript program that displays instructions based on traffic signal color.

📌 Requirements

- Store signal color in a variable ("red", "yellow", "green")
- Use a **switch statement** to display:
- Red → Stop
- Yellow → Get Ready

- Green → Go

Handle invalid signal input gracefully

Technical Constraints

- Must use switch–case
- Use string data types
- Use console.log() for output
- No if–else allowed

Code:

```
<!DOCTYPE html>

<html>

<head>

  <title>Traffic Signal Simulator</title>

</head>

<body>

  <h1>Traffic Signal</h1>

  <script>

    // Store signal color

    let signal =prompt("Enter signal");

    // Use switch statement

    switch (signal) {

      case "red":

        console.log("Stop");

        break;

      case "yellow":

        console.log("Get Ready");

        break;
```

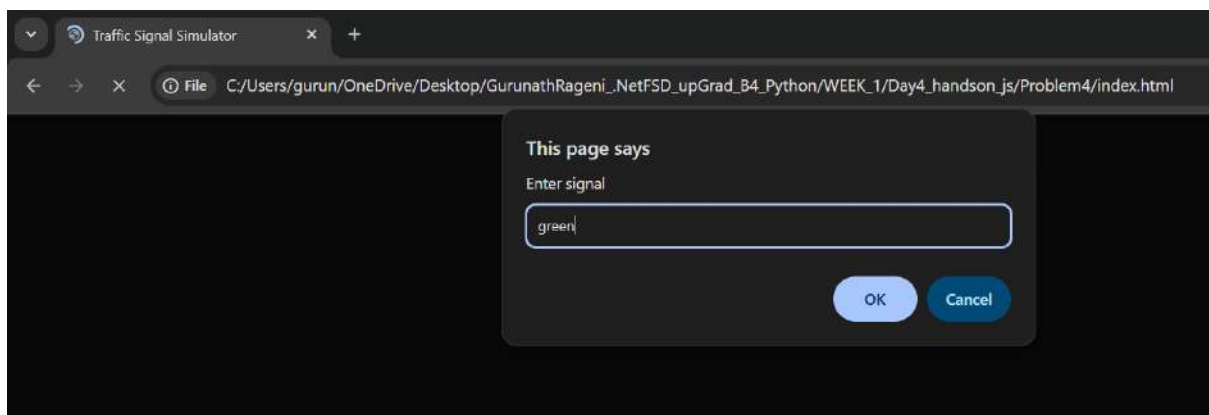
```
    case "green":
        console.log("Go");
        break;
    default:
        console.log("Invalid Signal Color");
}
</script>

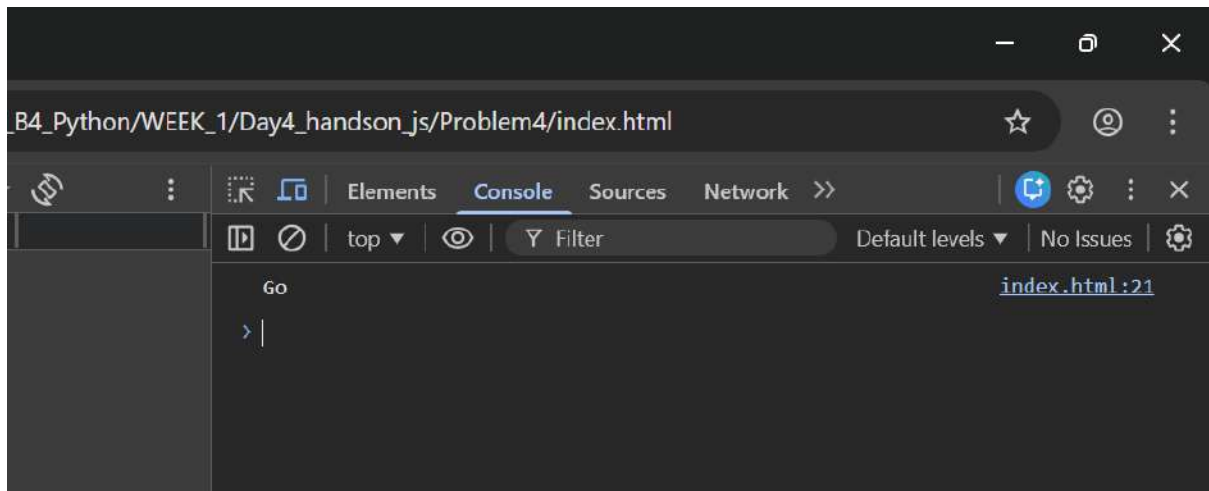
</body>
</html>
```

Code screenshot:

```
<> index.html M X
WEEK_1 > Day4_handson_js > Problem4 > <> index.html > html > body > script
1  <!DOCTYPE html>
2  <html>
3  <head>
4  |   <title>Traffic Signal Simulator</title>
5  </head>
6  <body>
7  |   <h1>Traffic Signal</h1>
8  |
9  <script>
10 |   // Store signal color
11 |   let signal =prompt("Enter signal");
12 |   // Use switch statement
13 |   switch (signal) {
14 |       case "red":
15 |           console.log("Stop");
16 |           break;
17 |       case "yellow":
18 |           console.log("Get Ready");
19 |           break;
20 |       case "green":
21 |           console.log("Go");
22 |           break;
23 |       default:
24 |           console.log("Invalid Signal Color");
25 |   }
26 </script>
27
28 </body>
29 </html>
```

Output Screenshot:





Explanation:

In this Problem, I created a Traffic Signal Simulator using HTML and JavaScript to demonstrate the use of the switch statement. The program takes user input through the prompt() function to enter a traffic signal color. Based on the entered value (red, yellow, or green), the switch case executes the corresponding action such as “Stop,” “Get Ready,” or “Go.” If the user enters an invalid color, the default case handles it by displaying an appropriate message. This project demonstrates decision-making control flow using the switch statement in JavaScript.

Problem 5: Number Analysis Tool (Level-2)

Scenario

A utility program is required to analyze numbers and provide insights such as positivity, parity, and range.

🔴 Requirements

- Store a number in a variable
- Use **conditional (ternary) operator** to check:
- Positive or Negative
- Use **if-else** to check:
- Even or Odd
- Use a **loop** to print all numbers from 1 to the given number

🔧 Technical Constraints

- Store a number in a variable
- Use conditional (ternary) operator to check:
- Positive or Negative
- Use if-else to check:

- Even or Odd
- Use a loop to print all numbers from 1 to the given number

Code:

```
<!DOCTYPE html>

<html>

<head>

  <title>Number Analysis Tool</title>

</head>

<body>

  <h1>Number Analysis</h1>

<script>

  // Store number in a variable

  let number = 10;

  let result = (number > 0) ? "Positive" : "Negative";

  console.log("Number Type: " + result);

  if (number % 2 === 0) {

    console.log("Parity: Even Number");

  } else {

    console.log("Parity: Odd Number");

  }

  console.log("Numbers from 1 to " + number + ":");

  for (let i = 1; i <= number; i++) {

    console.log(i);

  }

</script>

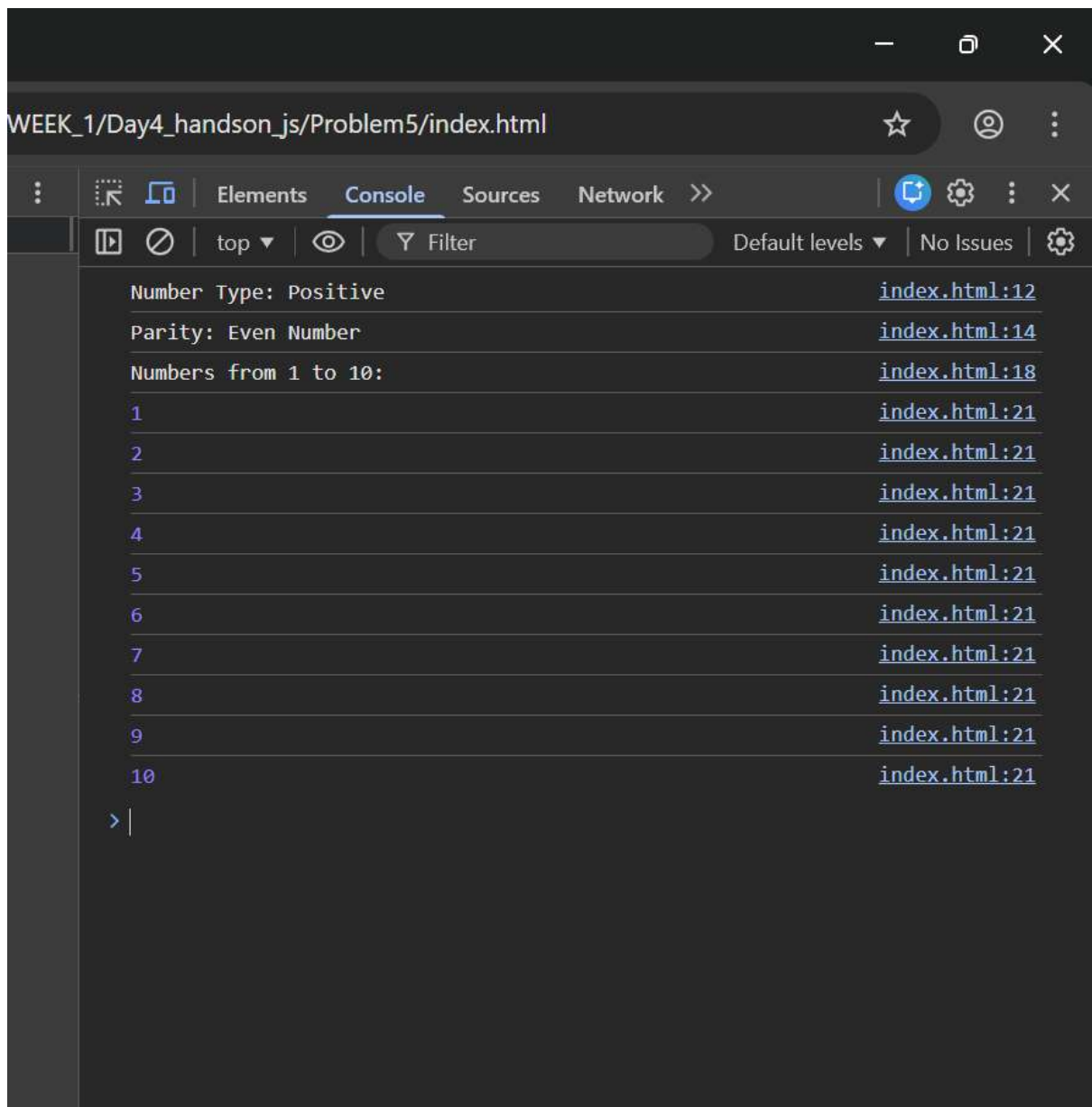
</body>

</html>
```

Code Screenshot:

```
<> index.html M X
WEEK_1 > C:\Users\gurun\OneDrive\Desktop\GurunathRageni_NetFSD_upGrad_B4_Python\WEEK_1\Day
1 < 4_handson_js\Problem5\index.html • Modified
2 <html>
3 <head>
4   <title>Number Analysis Tool</title>
5 </head>
6 <body>
7   <h1>Number Analysis</h1>
8 <script>
9   // Store number in a variable
10  let number = 10;
11  let result = (number > 0) ? "Positive" : "Negative";
12  console.log("Number Type: " + result);
13  if (number % 2 === 0) {
14    console.log("Parity: Even Number");
15  } else {
16    console.log("Parity: Odd Number");
17  }
18  console.log("Numbers from 1 to " + number + ":");
19
20  for (let i = 1; i <= number; i++) {
21    console.log(i);
22  }
23 </script>
24
25 </body>
26 </html>
```

Code Output:



Explanation:

In this Project, I developed a Number Analysis Tool using HTML and JavaScript to perform multiple operations on a given number. The program uses the ternary operator to determine whether the number is positive or negative. It then applies an if-else condition to check whether the number is even or odd using the modulus operator. Additionally, a for loop is implemented to print numbers from 1 up to the given number. This project demonstrates the use of operators, conditional statements, and looping structures in JavaScript for basic number analysis.